

View results

Respondent

60

Anonymous

132:01

Time to complete

A. General information about the respondent

1. * Country

Slovakia



2. * Name of organization(s) or entity(s) responsible for the preparation of the report

The Slovak Centre for Scientific and Technical Information

3. * Website of organization(s) or entity(s) responsible for the preparation of the report

<https://otvorenaveda.cvtisr.sk/>

4. * Email address of organization(s) or entity(s) responsible for the preparation of the report

5. Phone number of organization(s) or entity(s) responsible for the preparation of the report

6. Please describe the role/mandate of your organization

The national coordinator for research development and innovations

7. * Full name of officially designated contact person(s) who completed this survey

Jitka Dobbersteinová

8. * Position of officially designated contact person(s) who completed this survey

Open Science Departement Manager

9. * Email address of officially designated contact person(s) who completed this survey

10. Full Name(s) of designated official(s) certifying the report

Robert Gálik

11. Brief description of the consultation process established for the preparation of the report

The survey was consulted with members of the Working Group for Open Science operating under the Ministry of Education, Research, Development, and Youth of the Slovak Republic. The members of the Working Group are the elected representatives of the Slovak academic and scientific community, including learning societies.

12. * Name of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

The Slovak Academy of Sciences

13. * Website of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey
(Please use commas to separate multiple links, as: link A, link B)

<https://www.sav.sk/>

14. * Email address of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

15. Sector of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

☒

Public

☐

Private

☐

Civil Society

☐

Other

B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING

ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

Part 1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE,

ASSOCIATED BENEFITS AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN
SCIENCE

16. 1.1 Has the 2021 UNESCO Recommendation on Open Science been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

☒

Yes

☐

No

17. 1.1 cont. **If yes**, please indicate which ministries and or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

the Ministry of Education, Research, Development, and Youth, the Slovak Academy of Sciences, Slovak Rectors' Conference

18. 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

☐ Yes

☒ No

19. 1.3 Have there been awareness raising activities on open science, including all its key elements[i], associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (*ref.: (i) 16.f, 16.g, 16.h*)

[i] Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.

☒ Yes

☐ No

20. 1.3 cont. **If yes**, please provide details on the activities organized or foreseen and links as relevant.

All virtual educational activities here: <https://otvorenaveda.cvtisr.sk/category/webinare/>, our open science courses here: <https://otvorenaveda.cvtisr.sk/category/kurzy/> and open science events here: <https://otvorenaveda.cvtisr.sk/category/podujatia/>

21. 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science[i], in publicly funded research? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.

Please see the values of open science, namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.

Please see the guiding principles for open science ,namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

☒ Yes

☐ No

22. 1.4 cont. If **yes**, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

In 2024 the government accepted Code of Research Ethics and Integrity in Slovakia which implements open and transparent access to research publications and data as one of the principles (<https://vaia.gov.sk/wp-content/uploads/2024/07/Bod-2.-KODEX-vyskumnej-etiky-a-integrity-2024.pdf>), in 2024 the Slovak government accepted the National Action Plan for Open Science which defines the key activities of open science in Slovakia that were implemented in 2024 and are continuing in 2025: <https://otvorenaveda.cvtisr.sk/narodna-strategia-otvorenej-vedy/>

Part 2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

23. 2.1 Does your country have a national policy[i], strategy or plan of action on science, technology and innovation (STI)?

[i] National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.

☒ Yes

☐ No

24. 2.1 cont. **If yes**

- please share a machine-readable PDF file of the policy using this link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*

In the comment box, please indicate the question number for this attachment / document.

After uploading, please return to this page to continue the survey.

for Slovakia sent 17/03/2025

25. 2.1 cont. **If yes:**

- does it include any of the following open science elements:

- ☒ open access to scientific knowledge
- ☒ open science infrastructure
- ☒ open engagement of societal actors
- ☐ open dialogue with other knowledge systems

26. 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)[i] at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

- ☒ Yes
- ☐ Partly
- ☐ No

27. 2.2 cont. **If yes or partly**, please share a machine-readable PDF file(s) of the policy(ies) using this link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information for each: *Title of the policy or legal instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.*

the same as from 23 to 25 question

28. 2.3 In your country, are there specific policy instruments[i] that aim to promote open science?

[i] Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).



Yes



Under development



No

29. 2.3 cont. **If yes or under development**, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

The Contact Office of Open Access was established at the Slovak Centre of Scientific and Technical Information (the SCSTI) in 2016. It plays a key role as a national coordinator and methodological body for open science in Slovakia. One of the activities of the office is to raise awareness about open access and open science and fulfil the tasks arising from international projects in which the SCSTI participates. The office actively participates in fulfilling the tasks of the National Strategy for Open Science for 2021-2028 and the Action Plan to the National Strategy (updated every 2 years) and supports collaboration between individual stakeholders of open science and open access publishing in Slovakia. Every year, the Contact Office for Open Access organizes professional events, webinars, accredited training and courses to support open science, and provides consultation services.

30. 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)

- ☒ Yes
- ☐ Under development
- ☐ No

31. 2.4 cont. **If yes or under development**, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

The Action Plan for Open Science 2024 in Slovakia represents the second phase in implementing the National Strategy for Open Science 2021–2028. Approved by the Slovak Government in February 2024, this plan aims to further integrate open science principles into the country's research and development sectors, aligning with the long-term priorities outlined in the UNESCO Recommendations.

32. 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: (ii) 17.d, 17.e, 17.g)

- ☐ Yes
- ☒ Under development
- ☐ No

33. 2.6 In your country, are there specific funding mechanisms[i] for open science at the national level? (ref.: (ii) 17.j, (iii) 18.j, (iv) 19.c)

[i] A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.



Yes



Under development



No

34. 2.6 cont. **If yes or under development**, please provide details and links as relevant.

Currently there are three long-term transformative agreements with three publishers (Elsevier, IEEE, SpringerNature) covering gold open access publishing with these publishers.

35. 2.7 1.1 In your country, have open science practices[i] been integrated in existing research funding mechanisms?

[i] Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.

☐ Yes

☒ No

36. 2.7 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Difficult and fragmented organizational structure of grant agencies, where each organization has a different system and different priorities.

37. 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17.i)

☒ Yes

☐ No

38. 2.8 cont. **If yes**, please provide details and links as relevant.

The Action Plan for Open Science 2025 - 2028 as the national implementation road to openness is under development

39. 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: (ii) 17.j, 23.c)

- ☐ Yes
- ☒ Under development
- ☐ No

40. 2.9 cont. **If yes or under development**, please provide details and links as relevant.

Monitoring of open access publication is under development, links are not available yet

Part 3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

41. 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

In 2022, Slovakia's expenditure on research and development (R&D) was approximately 0.84% of its GDP. <https://ec.europa.eu/eurostat/web/products-eurostat-news/w/ddn-20231201-2>

42. 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

As of 2023, Slovakia's internet landscape is characterized by high penetration rates and substantial broadband access: internet Penetration: Approximately 91.2% of the Slovak population. In 2021, 89.93% of households had broadband internet access, reflecting a continuous upward trend in broadband adoption. <https://www.statista.com/statistics/702539/broadband-internet-household-penetration-slovakia/>

43. 3.3 Are there national research and education networks (NRENs)[i] active in the country? (ref.: (iii) 18.c)

[i] A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

- ☒ Yes
- ☐ No

44. 3.3 cont. **If yes**, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

Slovakia has an active National Research and Education Network (NREN) known as SANET (Slovak Academic Network). It is a member of the GÉANT network, connects universities, research institutions, and other educational institutions.

45. 3.4 Are there national or institutional open science infrastructures[i] in your country? (ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)

[i] Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.



Yes



Under development



No

46. 3.4 cont. **If yes or under development**, please indicate which type(s) of infrastructure:



federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage



community managed infrastructures, protocols and standards, for example those that support bibliodiversity and engagement with society



platforms for exchanges and co-creation of knowledge between scientists and society



community-based monitoring and information systems to complement national, regional and global data and information systems

47. 3.4 cont. **If yes or under development**, for each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

Slovak Centre of Scientific and Technical Information (CVTI SR) serves as the national information center ensuring the operation of interdisciplinary R&D centers: <https://www.vedatechnika.sk/sk/>. SCSTI is also a member of EOSC Association. Slovak Academic Network (SANET) is the national research and education networking organization connecting locations in 23 towns and provides high-speed internet connectivity to universities, research institutions,

48. 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 18.f)

☐ Yes

☒ No

49. 3.5. cont. **If no**, are the existing open science infrastructures at regional and international levels accessible to all the relevant open science actors from your country?

☒ Yes

☐ No

☐ Other

50. 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g)

☒ Yes

☐ No

51. 3.6 cont. **If yes**, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science, stage of research, accessibility to all the relevant users in the region or internationally).

This year we joined the pilot phase of establishing EOSC federation - Slovakia is EOSC node. <https://open-science-cloud.ec.europa.eu/>. EOSC EU Node is the first European-level node of the emerging EOSC Federation, which is envisioned to be a network of interconnected autonomous nodes, all operating under a common framework of best practices, including open science principles.

Part 4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY

AND CAPACITY BUILDING FOR OPEN SCIENCE

52. 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c)

☐ Yes

☒ No

53. 4.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Currently, there is no systematic education in the field of open science at the higher education level, primarily due to the fragmentation of the system. In Slovakia, this education is addressed through train-the-trainer courses at the level of further education.

54. 4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

☐ Yes

☒ No

55. 4.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Capacity building programmes for policy makers are not relevant at the moment, knowledge building is primarily done through the Working Group for Open Science at the Ministry of Education, which includes representatives of government authorities, the academic community, and learning societies.

56. 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: (iv) 19.b)

☐ Yes

☒ No

57. 4.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

The same as in the question 52.

58. 4.4 Are open educational resources[i] as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: (iv) 19.d)

[i] Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others (2019 Recommendation on Open Educational Resources)

- ☐ Yes
- ☐ Partly
- ☒ No

59. 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: (iv) 19.e)

- ☒ Yes
- ☐ No

60. 4.5 cont. **If yes**, please provide more information and links as relevant.

Slovak Reproducibility Network: a network of researchers focusing on the implementation of the transparency of research (the philosophy of open science) and the implementation of current methodological and statistical research practices. <https://www.slovakrn.org/>

Part 5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

61. 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: (v) 20.b, 20c, 20i)

☐ Yes

☒ No

62. 5.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

In Slovakia there is a shift to peer-review evaluation of research publication under development right now.

63. 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: (v) 20.a, 20.c, 20.d)

☐ Yes

☒ No

64. 5.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

65. 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h)

☐ Yes

☒ No

66. 5.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

67. 5.4 Have there been any unintended negative consequences[i] of open science practices reported in your country? (ref.: (v) 20)

[i] For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.

☒ Yes

☐ No

68. 5.4 cont. **If yes**, please provide more information and specify if there is a strategy or plan to preventing and mitigating such consequences.

High article processing charges associated with gold OA business models in scientific publishing

Part 6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES

OF THE SCIENTIFIC PROCESS

69. 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: (vi) 21)

- ☐ YES, at the national level
- ☐ YES, at the institutional level
- ☐ YES, at both the national and institutional levels
- ☒ No

70. 6.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Part 7. PROMOTING INTERNATIONAL AND MULTI- STAKEHOLDER COOPERATION IN THE CONTEXT

OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

71. 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

Yes, Slovakia has established a comprehensive framework to promote cross-border collaboration: participation in the European Open Science Cloud (EOSC), partnering with OpenAIRE or Slovak research institutions actively engage in cross-border projects, such as the CAPSID project:

72. 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e, 22.f)

☐ Yes

☒ No

73. 7.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

74. 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: (vii) 22.c)

☐ Yes

☒ No

75. 7.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Part 8. GENERAL CONSIDERATIONS

76. 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?



Yes



No

77. 8.1 cont. **If yes**, please provide more information.

Potencial shortage of research funding

78. ANY OTHER COMMENTS: If you wish, you may enter any additional comments or information here. You may also contact openscience@unesco.org